

Smrithi Lokesh Seramalanna

Email | Google Scholar | LinkedIn | Personal Website

Education

University of Pennsylvania, MSE Robotics

Expected: December 2027

Birla Institute of Technology and Science, Pilani, B.E. (Hons.) Mechanical Engineering, Minor in Robotics and Automation – CGPA: 9.14/10.00

July 2025

Experience

Research Assistant, Janus Intelligent Robot Lab, GRASP, University of Pennsylvania – Philadelphia, Pennsylvania

May 2026 – Present

- Developing visuomotor behavioral-cloning pipelines for bimanual manipulation on UR5 robots, using wrist-camera observations and diffusion policies to automate multi-step chemistry laboratory tasks.
- Curating multimodal demonstration datasets, filtering low-information plateaus, incorporating action history and DAgger corrections, and investigating joint training across multiple task policies to improve execution success.
- Extending the system to long-horizon manipulation tasks while debugging sim-to-real and hardware failures and scaling data collection and policy training across tasks; research in progress toward an ICRA 2027 submission.

Robotics Engineer, Open Science Stack – Bengaluru, India

Aug 2025 – Dec 2025

- Developed and deployed a full autonomy stack for and xArm robots for a autonomous pharma-lab-equivalent setup.
- Migrated the robotics stack from ROS to ROS 2, switched simulation environment to IsaacSim and implemented GPU-accelerated motion planning using cuMotion integrated with MoveIt2.

Exchange Student, Mälardalens University – Västerås, Sweden

Jan 2025 - Jun 2025

- Recipient of European Union's Erasmus+ ICM Scholarship enrolled in advanced robotics coursework specializing in planning, perception and autonomous systems

Research Intern, Carnegie Mellon University – Pittsburgh, USA

Jul 2024 - Jan 2025

- Developed a hierarchical planner for the Cuboid Multi-Agent Collective Construction (Cuboid-MACC) problem.
- Used graph-search based approaches to determine sequences of abstract block placements/removals, ensuring executability by verifying reachable access poses; dependencies between actions stored as an acyclic Action Dependency Graph.
- Extended Conflict-Based Search (CBS) to resolve multi-robot path conflicts in dynamic environments accounting for the rotational asymmetry of cuboid blocks and carried-block collision constraints; validated on 200 randomly generated structures.

Research Intern, Indian Institute of Science x Cornell University – Bangalore, India

May 2024 - Jul 2024

- Researched soft robot stiffness in the Ruina Lab (Prof. Andy Ruina), designing and fabricating fiber-reinforced, pneumatically actuated soft bodies.
- Engineered an experimental setup to analyze how transmitted forces vary under controlled pressure, volume, and control-cable tensioning.

Research Intern, Defence Research and Development Organisation – Hyderabad, India

May 2023 - Jul 2023

- Research in Advanced Systems Laboratory under Dr Lokesh Srivastava, Scientist 'F' on experimental characterization of carbon-epoxy composites for rocket motor casings and reliability prediction of composite samples by linear curve-fitting and Weibull analysis.

Publications

Hierarchical Planning for the Cuboid Multi-Agent Collective Construction (Cuboid-MACC) Problem

Jun 2026

Junil Min, Shambhavi Singh, Geordan Gutow, **Smrithi Lokesh Seramalanna**, Bhaskar Vundurthy, Jinwoo Choi, Howie Choset

Accepted to the 2026 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)

Kinematic models for Cable-driven Continuum Robots with multiple segments and varying cable offsets

Sept 2024

Ashish Bhalkikar, **Smrithi Lokesh**, Ashwin K P

doi.org/10.1016/j.mechmachtheory.2024.105701

Projects

Autonomous VIO-Based Quadcopter Navigation – MEAM 6200 Advanced Robotics Spring 2026

- Integrated A* planning on a voxelized map, minimum-snap trajectory generation, SE(3) geometric control, and an Error-State Kalman Filter (ESKF) for IMU/stereo-visual fusion into a full closed-loop autonomy stack.
- Achieved safe navigation across six obstacle environments (maze, slalom, stairwell, switchback, window, over/under) using only on-board VIO state estimates; co-designed trajectory aggressiveness with estimator bandwidth to prevent visual feature dropout.

Two-View Stereo 3D Reconstruction – CIS 5800 Machine Perception Spring 2026

- Implemented a complete stereo pipeline: rectification, disparity map computation (SSD / SAD /ZNCC with L-R consistency), depth estimation, and back-projected point-cloud generation in world frame.
- Aggregated multi-pair reconstructions to produce dense 3D point clouds of real scenes.

Aircraft Design and Analysis September 2022 - April 2024

- Designed and built a micro-class fixed wing aircraft to demonstrate Urban Air Mobility (UAM) missions for AIAA's Design, Build and Fly 2024.

Software for Machine Component Design January 2024 - May 2024

- Developed a repository aimed at complementing the design process detailed in Shigley's Mechanical Engineering Design in collaboration with the Teaching Learning Centre at BITS Pilani. The repository served as a resource hub for mechanical engineering students and professionals.

Technical Skills

Programming Languages: Python, C++ , Java, MATLAB/Simulink

Robotics & Simulation: ROS 2, MoveIt2, cuMotion, NVIDIA Isaac Sim, Gazebo, RViz, PyTorch, OpenCV

Hardware: UR3e, UR5, Franka Emika Panda, xArm, Robotiq Grippers, 3D Printing, Laser Cutting, CNC, Arduino, Raspberry Pi

CAD & Design: SolidWorks, Fusion 360, XFLR5

Selected Undergraduate Coursework: BITS F441 Robotics, EEE F242 Control Systems, ME F320 Engineering Optimization, EEE G512 Embedded System Design, ME F319 Vibrations and Control

Graduate Coursework: CIS 5200 Machine Learning, CIS 5800 Machine Perception, MEAM 6200 Advanced Robotics

Teaching and Volunteering

Lab Staff, Additive Manufacturing Lab (AddLab), University of Pennsylvania May 2026 – Present

- Support research labs and student engineering projects through additive manufacturing, rapid prototyping, and fabrication.
- Operate and maintain industrial and desktop 3D printing systems, including FDM, SLA, and resin-based technologies.

Teaching Assistant – Artificial Intelligence, Engineering Summer Academy, University of Pennsylvania July 2026

- Working with Prof. Robert Ghrist, supporting high-school students in a summer program on foundational artificial intelligence and algorithmic thinking.

Scratch Instructor, Fife-Penn STEM and CS Academy – St. Francis Xavier School January 2026 – Present

- Teaching foundational programming concepts to K–8 students to prepare students for higher-level STEM education.

Undergraduate Teaching Assistant – Mechanics of Solids, BITS Pilani Aug 2023 – Dec 2023

- TA under Prof. Kiran Dinkar Mali; addressed student queries, evaluated quizzes, and managed academic records for a class of 180 sophomore students.

Instructor – Aircraft Design Course, Centre of Technical Education, BITS Pilani Nov 2022 – Jan 2023

- Conducted a short course on UAV performance and design open to all freshmen, covering airfoil and wing analysis, design parameter optimisation and CAD.

Awards and Achievements

Erasmus+ ICM Scholarship: European Union

Overall Excellence Award: International Rover Challenge 2023

26/107 International Rank: AIAA Design, Build and Fly 2024

Partial Travel Grant: International Programs and Collaborations Division, BITS Pilani